INCH-POUND

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DETAIL SPECIFICATION

GENERATOR, GAS, DRY BAY FIRE EXTINGUISHER, GGU-15A/A

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Dry Bay Fire Extinguisher Gas Generator, GGU-15A/A, referred to herein as "gas generator."

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202 - Electronic and Electrical Component Parts

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS (CAGE CODE 30003)

838AS243 - Generator, Gas, Dry Bay Fire Extinguisher GGU-15A/A

838AS254 - Cover, Electrical Connector Gas Generator Assembly (for Helium Leak Test)

3205AS191 - Test Fixture Assembly (100 ± 0.5 Liter)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3205AS199 - Container, Shipping and Storage, Gas Generator

3205AS216 - Vibration/Shock Plate and Gas Generator Assembly

(Copies of these drawings are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D1141	-	Standard Practice for Preparation of Substitute Ocean Water
ASTM E1742/E1742M	-	Standard Practice for Radiographic Examination

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.1.

3.2 Flight critical part. The gas generator is a flight critical item, and the technical data package shall be identified and procured as such (see 6.4.3).

3.3 Design and construction. The GGU-15A/A gas generator covered by this specification shall be manufactured in accordance with Drawing 838AS243.

3.3.1 Materials. All materials used in the manufacture of the gas generator shall conform to the standards established at technical service release as specified (see 6.2). Any divergence shall require technical evaluation and potential design certification testing.

3.4 Primary components. Ignition pellets, propellant pellets, and the electrical pressure cartridge are the gas generator primary components (see 6.4.5). Only one lot of each primary component shall be used in a gas generator lot. One primary lot may be used in more than one gas generator lot.

3.4.1 Electrical pressure cartridge. The zirconium potassium perchlorate ignition charge and the titanium hydride/potassium perchlorate main charge are the electrical pressure cartridge primary components. Only one lot of each primary component shall be used in an electrical pressure cartridge lot. One primary lot may be used in more than one electrical pressure cartridge lot.

3.5 Gas generator production.

3.5.1 Visual requirement. The gas generators shall conform to Drawing 838AS243 and be free of the following defects: illegible, missing, or inaccurate markings; burrs, dents, deep scratches, cracks, corrosion, sharp edges, bent or missing electrical contact pins; damaged mounting studs; chipped or incomplete protective finish; or other defects that may prevent the installation of the gas generator.

3.5.1.1 Shorting plug. The electrical pressure cartridge shall have a shorting plug (see 6.4.1) installed that shorts all pins.

3.5.1.2 Protective caps. Protective caps (see 6.4.1) shall be installed on the mounting studs.

3.5.2 Radiographic examination requirement. When radiographically examined in accordance with 4.3.2, the gas generators shall show proper assembly, presence of parts, sealing, and presence of primary components. The radiographic inspection shall show no foreign materials present.

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3.5.3 Leakage. Gas generators shall not exhibit a leak rate in excess of 2.7 x 10^{-5} cubic centimeters per second (cm^3/s) of helium (1.1 x 10^{-5} cm^3/s of air) at a pressure differential of 1.0±0.1 atmospheres when tested in accordance with 4.3.3.

3.5.4 Bridgewire resistance. The bridgewire resistance between pins A - B and C - D shall be 0.90 to 1.10 ohms and the shunt bridgewire resistance between pins B - C shall be 0.70 to 1.30 ohms when tested in accordance with 4.3.4.

3.6 Gas generator performance. Gas generators shall meet all requirements of 3.6.1 through 3.6.3.

3.6.1 Electrical characteristics.

3.6.1.1 Electrostatic discharge. Gas generators shall not fire, dud, or otherwise be affected when tested in accordance with 4.3.5.

3.6.1.2 Dielectric withstanding voltage. Gas generators shall not exhibit a leakage current in excess of 0.1 milliampere when tested in accordance with 4.3.6.

3.6.1.3 Stray voltage. Gas generators shall not fire, dud, or otherwise be affected when tested in accordance with 4.3.7.

3.6.2 Environmental.

3.6.2.1 Six-foot drop. Gas generators shall not fire or incur any detrimental internal/external damage or degradation as a result of the six-foot drop test in accordance with 4.3.8 and shall meet the functioning requirements of 3.6.3 when tested.

3.6.2.2 Shock. Gas generators shall not initiate or incur any internal/external damage or degradation when shocks are applied in accordance with 4.3.9 and shall meet the functioning requirements of 3.6.3 when tested.

3.6.2.3 Vibration. Gas generators shall not initiate or incur any internal/external damage or degradation when vibration cycles are applied in accordance with 4.3.10 and shall meet the functioning requirements of 3.6.3 when tested.

3.6.2.4 Temperature/humidity/altitude (TH&A) cycling. Gas generators shall not initiate or incur any internal/external damage or degradation when subjected to TH&A cycling in accordance with 4.3.11 and shall meet the functioning requirements of 3.6.3 when tested.

3.6.2.5 Salt fog. Gas generators shall not initiate or incur any internal/external damage or degradation when subjected to salt fog testing in accordance with 4.3.12 and shall meet the functioning requirements of 3.6.3 when tested.

3.6.3 Functioning. Gas generators shall meet the requirements of 3.6.3.1 through 3.6.3.4 when fired in accordance with 4.3.13.

3.6.3.1 Maximum pressure. The upper and lower limits for maximum pressure (in pounds per square inch absolute [psia]) when functioned over a temperature range from -72 to +160 °F shall be 50 psia and 20 psia, respectively.

3.6.3.2 Time to 95 percent of maximum pressure. The limit for time to 95 percent of maximum pressure shall not exceed 200 milliseconds (see 6.4.7).

3.6.3.3 Ignition delay. The ignition delay shall not exceed 25 milliseconds (see 6.4.4).

3.6.3.4 Delta (Δ) weight. The change in the weight of the gas generator, following functioning, shall not be less than 100 grams (see 6.4.2).

3.7 Workmanship. The gas generators shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes, sealing, and assembly operations when inspected in accordance with 4.3.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

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b. Production inspection (see 4.2.2).

c. Lot acceptance inspection (see 4.2.3).

4.2 Types of inspections.

4.2.1 First article inspection. Unless otherwise specified (see 6.2), a first article inspection sample of 27 gas generators conforming to Drawing 838AS243 shall be subjected to first article testing. Twenty-four of these gas generators shall be expended in the tests listed in table I and three shall be retained for investigative purposes. Any damage inflicted by the environmental exposures that would adversely affect the performance of the item in service application shall be cause for rejection of the first article sample. Any production prior to notification by the contracting activity of the first article sample acceptability is at the contractor's risk. First article testing is for the purpose of determining that the contractor, prior to starting production, is capable to producing items that comply with the technical requirements as specified herein. A first article inspection shall be required if two or more years have passed since the item was manufactured, after major turnover in personnel, following major changes to the production facility, or following transfer of production to a new facility or site. Failure of any gas generator to comply with the requirements of table I shall be cause for rejection of the first article represented.

TABLE I. First article test plan.

Test sequence	Test paragraph	Requirement paragraph	Sample groups					
			I	II	III	IV	V	Total
1. Visual inspection	4.3.1	3.4, 3.5.1, 3.7	6	3	6	9	3	27
2. Radiographic examination	4.3.2	3.5.2	6	3	6	9	3	27
3. Leakage	4.3.3	3.5.3	6	3	6	9	3	27
4. Bridgewire resistance	4.3.4	3.5.4	6	3	6	9	3	27
5. Electrostatic discharge	4.3.5	3.6.1.1	6	3	6	9	---	24
6. Bridgewire resistance	4.3.4	3.5.4	6	3	6	9	---	24
7. Dielectric withstanding voltage	4.3.6	3.6.1.2	6	3	6	9	---	24
8. Bridgewire resistance	4.3.4	3.5.4	6	3	6	9	---	24
9. Stray voltage	4.3.7	3.6.1.3	6	3	6	9	---	24
10. Bridgewire resistance	4.3.4	3.5.4	6	3	6	9	---	24
11. 6-foot drop	4.3.8	3.6.2.1	6	---	---	---	---	6
12. Shock	4.3.9	3.6.2.2	6	---	---	---	---	6
13. Vibration	4.3.10	3.6.2.3	6	3	---	---	---	9
14. Temp./humidity/altitude	4.3.11	3.6.2.4	---	3	---	---	---	3
15. Salt fog	4.3.12	3.6.2.5	---	---	6	---	---	6
16. Radiographic examination	4.3.2	3.5.2	6	3	---	---	---	9
17. Leakage	4.3.3	3.5.3	6	3	6	---	---	15
18. Bridgewire resistance	4.3.4	3.5.4	6	3	6	---	---	15
19. Function test:	4.3.13	3.6.3	---	---	---	---	---	---
-72±5 °F			2	---	2	3	---	7
+70±5 °F			2	3	2	3	---	10
+160±5 °F			2	---	2	3	---	7

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4.2.2 Production inspection. All production gas generators shall be inspected and screened for defects. Gas generators failing to meet the requirements listed in table II shall be rejected and removed from the lot.

TABLE II. Production inspection.

Test sequence	Test paragraph	Requirement paragraph
1. Visual inspection	4.3.1	3.4, 3.5.1, 3.7
2. Radiographic examination	4.3.2	3.5.2
3. Leakage	4.3.3	3.5.3
4. Bridgewire resistance	4.3.4	3.5.4

4.2.3 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests specified in table III. Failure of any sample gas generator to comply with the requirements listed in table III shall be cause for rejection of the lot represented.

TABLE III. Lot acceptance inspection and tests.

Inspection/test	Test paragraph	Requirement paragraph	Quantity
1. Visual inspection	4.3.1	3.4, 3.5.1, and 3.7	Test and retain sample
2. Radiographic examination	4.3.2	3.5.2	Test and retain sample
3. Leakage	4.3.3	3.5.3	Test and retain sample
4. Bridgewire resistance	4.3.4	3.5.4	Test and retain sample
5. Electrostatic discharge	4.3.5	3.6.1.1	Test and retain sample
6. Bridgewire resistance	4.3.4	3.5.4	Test and retain sample
7. Dielectric withstanding voltage	4.3.6	3.6.1.2	Test and retain sample
8. Bridgewire resistance	4.3.4	3.5.4	Test and retain sample
9. Function test:	4.3.13	3.6.3	---
-72±5 °F			1/3 test sample
+70±5 °F			1/3 test sample
+160±5 °F			1/3 test sample

4.2.3.1 Sample size. Unless otherwise specified (see 6.2), a random sample of gas generators from each production lot, including the samples retained for investigative purposes, shall be selected in accordance with table IV for lot acceptance inspection. Test sample gas generators and samples retained for investigative purposes shall not be applied as part of the quantity specified for delivery (see 6.2).

TABLE IV. Lot acceptance sampling.

Lot size	Test sample size	Retained sample size
1-50	12	3
51-200	15	3
201-500	21	3

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4.3 Inspections and tests.

4.3.1 Visual inspections. The external condition and appearance of the gas generators shall be determined by comparison to Drawing 838AS243. Specific inspections of the electrical connector, shunting plug, mounting studs, aluminum case, and identification label shall be conducted as a minimum. Each gas generator shall meet the requirements of 3.5.1 and 3.7.

4.3.2 Radiographic examination. Radiographic examination shall be conducted in accordance with ASTM E1742/E1742M. All gas generators shall be identified with serial numbers beginning with 001 prior to examination. The gas generators shall be arranged on boards or trays in consecutive order with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4- x 6-inch region maximum. The radiographic identification shall include the drawing number, the lot number, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the lot acceptance tests (see 6.2). Any observable imperfections as outlined in 3.5.2 shall be cause for rejection of the gas generator. Defective gas generators found during radiographic review are to be marked on the radiographic plate and removed from the production lot. Radiographic equipment settings shall be noted on the radiographic review sheets.

4.3.3 Leakage test. The gas generators shall be subjected to the following leakage test:

a. For production inspections, conduct leakage test after clear-coat application but prior to pyrotechnic tape assembly. Also, for lot acceptance or first article inspections, remove pyrotechnic tape from each unit prior to test. Use care to avoid any surface deformation when removing pyrotechnic tape. (Note: helium can be trapped under pyrotechnic tape and provide false leak test readings or anomalies when test is conducted.)

b. Remove any protective caps from the units and wipe exterior surfaces clean using a dry, lint free material.

c. Secure cap, P/N 838AS254, on gas generator electric initiator.

d. Verify leak detector is operational. Record leak detector make, model, range, serial number, accuracy, date of last calibration, and calibration due date.

e. Place test units in a pressure vessel. (Note: leak testing on test units shall occur within 15+1 minutes from the time the test unit is removed from the pressure vessel. Therefore, process only as many units as can be leak tested within 15+1 minutes of helium bombing. Any unit not tested within 15+1 minutes shall be re-pressurized.)

f. Slowly evacuate the pressure vessel. Evacuate to a vacuum greater than 26 inches of mercury and close vacuum valve.

g. Seal and slowly pressurize the vessel with helium to 7.35±1.00 psig. Hold at this pressure condition for 60 +1/-0 minutes.

h. Remove the test units from the pressure vessel and record the time.

i. Remove cap, P/N 838AS254, and remove any retained grease from units. Flush exterior of test units thoroughly with low-pressure air to remove possible trapped helium.

j. Perform helium leak test on the test unit using leak detector. Record time at which leak test is conducted. The leakage shall meet the requirements of 3.5.3.

4.3.4 Bridgewire resistance. The initiator bridgewire resistance shall be measured across pins A-B, C-D, and B-C at 70±18 °F. The test current shall not exceed 50 milliamperes. The bridgewire resistance shall meet the requirements of 3.5.4.

4.3.5 Electrostatic discharge. The gas generators shall be subjected to the following electrostatic discharge test:

a. Temperature condition gas generators at 70±5 °F for a minimum of 12 hours prior to conducting test.

b. Remove protective shorting plug from test unit just prior to test.

c. Discharge a 25,000±500 volts direct current (Vdc) from a 500±25 pico-farad capacitor through a 5,000±250 ohm resistor connected in a 5 micro-henry total inductance series circuit between the shunted pins and case of gas generator electric pressure cartridge. Maintain the series connection for 60 seconds. The gas generator shall meet the requirements of 3.6.1.1.

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4.3.6 Dielectric withstanding voltage. The test circuit shall be connected between the shorted electrical pins and the case of the gas generator electrical pressure cartridge. The gas generator electrical pressure cartridge shall be subjected to 500±25 Vdc for 60 seconds. The current shall be measured with an accuracy of 5 percent and conducted in accordance with MIL-STD-202, Method 301. The gas generator shall meet the requirements of 3.6.1.2.

4.3.7 Stray voltage. The gas generators shall be subjected to the following stray voltage test:

a. Temperature condition gas generators at 70±5 °F for a minimum of 12 hours prior to conducting test.

b. Verify test environment of 70±5 °F.

c. Remove protective shorting plug from test unit just prior to test.

d. Subject gas generator to 2,000 pulses of direct current (DC) between bridgewire circuit A-B and C-D. Each pulse shall be 300 milliseconds duration and the pulse rate shall be two per second. Each pulse shall have an amplitude of 100±5 milliamperes.

e. Failure of any gas generator to comply with the requirements of 3.6.1.3 shall be cause for rejection of the lot or first article or both.

4.3.8 Six-foot drop. The gas generators shall be subjected to the six-foot drop test as follows:

a. Remove any protective caps that may interfere with the drop test.

b. Out of the test group, select two units to impact the gas generator electric pressure cartridge down, two impacts electric pressure cartridge up, and two impacts with gas generator horizontal. Refer to figure 1 for impact orientations. (Note: each test unit shall only be dropped once.)

c. Attach test unit to guide wire to ensure proper impact orientation.

d. Release lowest point of the test unit in the drop orientation from a vertical height of 6 feet + 2 inches -0 inches above the 2-inch-thick steel target plate embedded in concrete. The gas generator shall meet the requirements of 3.6.2.1.

e. Failure of any gas generator to comply with the requirements of 3.6.2.1 shall be cause for rejection of the lot or first article or both.

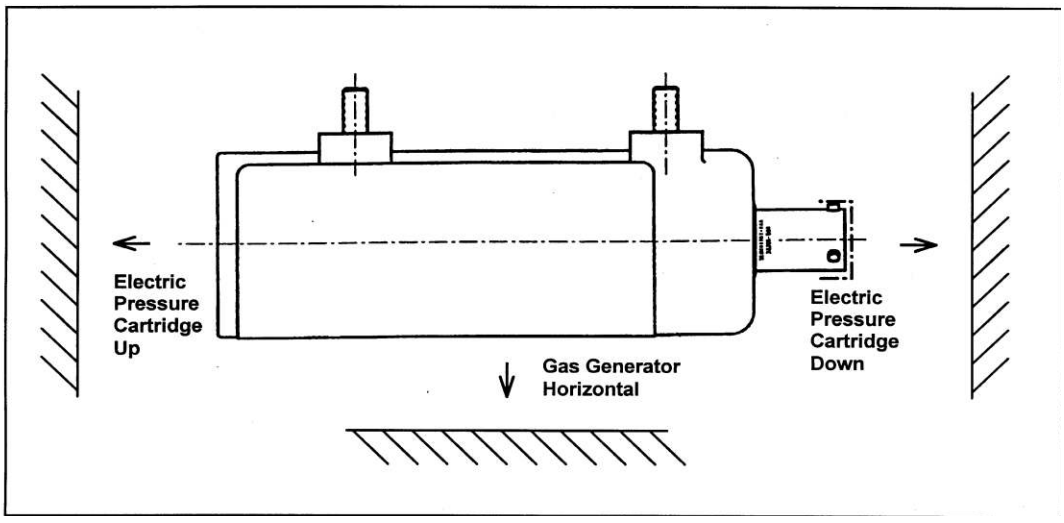


FIGURE 1. Six-foot drop orientations.

4.3.9 Shock. The gas generators shall be subjected to the shock drop test as follows:

a. Assemble test items on fixture in accordance with Drawing 3205AS216.

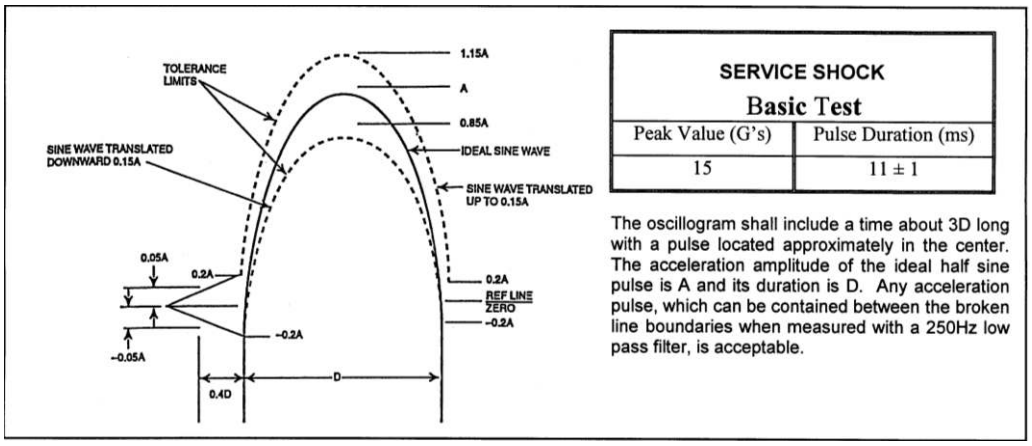
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b. Secure shock plate and gas generator assembly onto test table. An input control shock sensor shall be rigidly attached to the test fixture as near as possible to the gas generator attachment points. A response shock sensor shall be attached on the gas generator housing at the least supported location.

c. Verify protective shorting plug installed and sealed with electrical tape.

d. The test items shall be subjected to 18 impact shocks using the levels and durations specified on figure 2. Conduct shock testing at 70±18 °F temperature conditions. Apply impact shocks, three in each direction per orthogonal axes depicted on figure 3.

e. Failure of any gas generator to comply with the requirements of 3.6.2.2 shall be cause for rejection of the lot or first article or both.



SERVICE SHOCK Basic Test	
Peak Value (G's)	Pulse Duration (ms)
15	11 ± 1

The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. Any acceleration pulse, which can be contained between the broken line boundaries when measured with a 250Hz low pass filter, is acceptable.

FIGURE 2. Impact shock pulse.

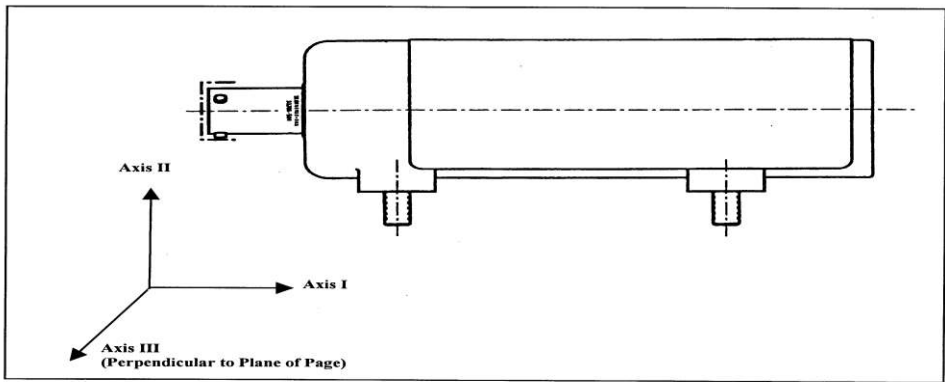


FIGURE 3. Shock and vibration axes.

4.3.10 Vibration. The gas generators shall be subjected to the vibration test as follows:

a. Remove protective caps from gas generator mounting studs.

b. Assemble test items on fixture in accordance with Drawing 3205AS216.

c. Secure vibration plate and gas generator assembly onto test table. An input control vibration sensor shall be rigidly attached to the test fixture as near as possible to the gas generator attachment points. A response vibration sensor shall be attached on the gas generator housing at the least supported location or a location shown to have significant amplification.

d. Verify protective shorting cap installed and sealed with electrical tape.

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e. A resonance survey is required to be performed. Perform a sine sweep from 5 hertz to 2,000 hertz at a frequency sweep rate of one octave/minute. The input level is .024-inch double amplitude or ±2 g's, whichever is less. Resonant points shall be noted, the response recorded and plotted, with the modes (see 6.4.6) of each resonance described. The survey shall be performed under the following conditions:

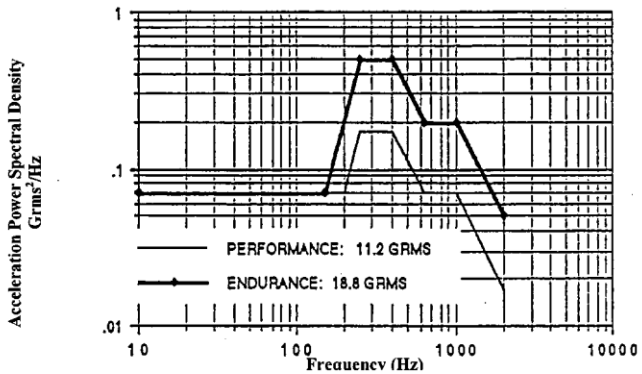
(1) at the beginning of demonstration testing for each axis of vibration;

(2) prior to restart of the test following modification or replacement of the vibration table, test fixture, or test item;

(3) if response data recorded during a particular vibration test indicates that the resonant frequencies have shifted in excess of ±10 percent.

f. Each test item shall be subjected to the figure 4 vibration schedule at -72 °F, +70 °F, and +160 °F. The vibration time per each axis shall be equally divided for vibration at -72 °F, +70 °F, and +160 °F. The gas generator axis orientation is shown on figure 3. The vibration fixture shall not have any structural resonance in the range of the vibration test frequencies.

g. Failure of any gas generator to comply with the requirements of 3.6.2.3 shall be cause for rejection of the lot or first article or both.



Frequency (Hz)	Performance Amplitude (G^2/Hz) or Slope	Endurance Amplitude (G^2/Hz) or Slope
10-151		0.07
10-200	0.07	
151-250		+12dB/Oct.
200-250	+ 12dB/Oct.	
250-400	0.17	0.5
400-633	- 6dB/Oct.	- 6dB/Oct.
633-1000	0.07	0.2
1000-2000	- 6dB/Oct.	- 6dB/Oct.

Random Test Schedule
(Test times in minutes per axis)

Random Performance	30
Random Endurance	120

FIGURE 4. Vibration profile.

4.3.11 TH&A cycling test. The gas generators shall be subjected to the TH&A cycling test, outlined in MIL-D-21625, with the following requirements:

a. Test to be conducted with protective cap sealed with tape and electrical bonding area covered with tape.

b. Place units on screen trays or in wire baskets to allow air to circulate freely, exposing all surfaces of the units, except noted otherwise.

c. The gas generators shall be subjected to a 28-day TH&A cycling test as indicated in table V. There shall be no interim withdrawals.

d. Failure of any gas generator to comply with the requirements of 3.6.2.4 shall be cause for rejection of the lot or first article or both.

4.3.12 Salt fog. The gas generators shall be subjected to the salt fog test as follows:

a. Test to be conducted with protective shorting cap installed and sealed with electrical tape and electrical bonding area (flat contact surface at mounting studs) also covered with electrical tape.

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TABLE V. Temperature/humidity/altitude cycling procedure.

DAY	TIME	PROCEDURE
MONDAY	0800	Place items in chamber maintained at 70 °F and 50% Relative Humidity (RH).
	1200	Raise chamber to 160 °F and 95% RH. Chamber shall reach conditions within one hour.
	1600	Remove items and immediately place in a chamber maintained at -72 °F at a pressure altitude (PA) of 60,000 feet (ft).
TUESDAY	0800	Remove items from chamber and place in chamber maintained at 70 °F and 50% RH.
	1200	Remove items from chamber and place in chamber maintained at -72 °F and 60,000 ft PA.
	1600	Remove items from chamber and place in chamber maintained at 160 °F and 95% RH.
WEDNESDAY	0800	Reduce chamber pressure to 70 °F at 50% RH. Chamber shall reach conditions within one hour.
	1200	Raise chamber to 160 °F and 95% RH. Chamber shall reach conditions in one hour.
	1600	Remove items and place in chamber maintained at -72 °F and 60,000 ft PA.
THURSDAY		Repeat Tuesday schedule.
FRIDAY	0800	Reduce chamber pressure to 70 °F at 50% RH. Chamber shall reach conditions within one hour.
	1200	Raise chamber to 160 °F and 95% RH. Chamber shall reach conditions in one hour.[1/]
	1600	Remove items and place in chamber maintained at -72 °F and ambient pressure.

FOOTNOTE:

[1/] Test schedule shall be repeated for a total of four weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 Friday until 0800 on Monday at 160 °F and 95% RH.

NOTES:

1. Tests shall be performed in the sequence shown, for durations and conditions shown. Day and clock times are provided as a guide and are not mandatory.
2. No interim withdrawals of items. All items submitted for test shall complete the entire 28-day test cycle.

b. The test shall be conducted in accordance with the procedure of ASTM B117 except that synthetic sea water conforming to ASTM D1141 shall be used in place of the 5 percent salt solution. The control of pH and salt impurity requirements per ASTM B117 shall not apply. Test duration shall be 168 hours. Waxed string shall be used to support the gas generators in the chamber. The cartridges shall be oriented horizontally with mounting studs down.

c. After completion of testing, the salt deposits (if any) shall be removed, if necessary, for aiding in a visual inspection. To remove the salt deposits, a gentle washing technique shall be used with water not warmer than 38 °C (100 °F). Then the gas generators shall be stored for 48 hours at 70 °F.

d. Failure of any gas generator to comply with the requirements of 3.6.2.5 shall be cause for rejection of the lot or first article or both.

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4.3.13 Functional performance. The gas generators shall be subjected to the functional test as follows:

a. Remove all protective caps except electric shunt and weigh each test item to the nearest gram.

b. Units shall be temperature conditioned for not less than 4 hours and no more than 24 hours at the specified temperatures.

c. Verify that the temperature of test fixture tank is at a temperature of 70±5 °F.

d. Test unit shall be functioned within 5 minutes maximum following removal from the conditioning box. If any test item is not fired within 5 minutes after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested.

e. Assemble test item on fixture in accordance with Drawing 3205AS191.

f. Evacuate the air from the test fixture to less than 0.5 psia.

g. Fire the test item by applying a 50 millisecond pulse of 3.5+0.5 ampere current across either bridgewire A-B or C-D. Approximately one-half (50 percent) of each group, submitted for functional test, shall be fired across bridge A-B and the other half across C-D.

h. Record the tank pressure versus time within the test fixture during test. The gas generators shall meet the requirements of 3.6.3.1 through 3.6.3.4.

4.3.13.1 Test failure. If a test failure is attributable to an assigned cause, excluding the gas generator, the test shall be declared a no-test and repeated.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gas generator is designed for use in the F/A-18E/F Aircraft Dry Bay Fire Suppression System. The Dry Bay Fire Suppression System is designed to protect the dry bays under fuel tanks two, three, and four. The gas generator is for military aircraft use and has no equivalent commercial application.

6.1.1 Operation. When the system is armed, all the gas generators are automatically discharged upon detection of a fire/explosion; no aircrew interaction is required. Six 33300-301 gas generators are installed in various locations within the dry bays. The gas generators are connected to a Fire Suppression Control Alarm (FSCA), which is connected to Optical Fire Detectors (OFD). Upon sensing a fire, the optical fire detectors transmit a signal to the FSCA. The FSCA is programmed to send electrical firing pulses in a pre-determined timing sequence to all gas generators. The gas generators, when initiated, the main propellant combustion produces inert gases comprised mainly of nitrogen, carbon dioxide, and water vapor. The gases vent radially through orifices in the body of the gas generators. These inert gases achieve fire extinguishment in the dry bays by reducing the oxygen level below that required for sustaining combustion. The gas is non-ozone depleting, non-toxic, and has a minimal corrosive effect on the aircraft dry bay structure. Once gas generators are fired, the system can only be restored to operational use by replacement of functioned units.

6.1.2 Changes to configuration. These gas generators were developed to function in the F/A-18E/F dry bays. Of major importance is the structural over pressure limits of the bays. Changes to the gas generators resulting in increased performance, faster burn times, or greater pressure output could result in the structural failure of the airframe components during an inadvertent actuation of the system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. That the safety precaution requirements of the "DoD Contractor's Safety Manual for Ammunition and Explosives," DoD Manual 4145.26 are applicable. (Note: when this specification is used as a part of the description of work to be accomplished by a Government activity, the safety precautions requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.)

c. Inspection conditions if other than as specified.

d. Whether a first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2.1).

e. Materials required as established by the technical service release (see 3.3.1).

f. Production lot size (quantity) and test facility (see 4.2.3.1).

g. Directions for shipping radiographs of entire production lot and marriage log (report indicating cartridge serial number to initiator serial number), along with the ballistic sample to the activity conducting the production lot acceptance tests (see 4.3.2).

h. Packaging requirements (see 5.1 and 6.5).

i. If bar coding is not required on outer container (see 6.5.2).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offeror(s) whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of the first articles. Invitation(s) for bid(s) should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Caps and plugs. Plastic shipping caps are installed on the end of the mounting studs and a shorting plug is installed on the electrical pressure cartridge.

6.4.2 Delta (Δ) weight. Weight of gas generator, without protective plugs and caps installed, minus the immediate post functioning weight of gas generator.

6.4.3 Flight critical part. A part, the single failure of which, during any operating condition, could cause the loss of aircraft or one of its major components, loss of control, or which may cause significant personnel injury, including during flight.

6.4.4 Ignition delay. Time from firing current application to first indication of pressure in test fixture.

6.4.5 Primary component. Primary components are all components in which a functional failure would result in a misfire or malfunction of the gas generator.

6.4.6 Resonant modes: A resonant mode is considered to be any frequency-dependent mechanical disturbance which can be detected visually, aurally or by means of other sensing devices. A resonant mode exists at any frequency at which the ratio of specimen response level to input level is at a peak such that both an increase and a decrease in the excitation frequency will produce a decrease in the specimen levels. A gain of two between the output and input generally constitutes a resonance.

6.4.7 Time to 95 percent of maximum pressure. Time from indication of pressure within the test fixture to when 95 percent of the maximum pressure is observed.

6.5 Contract packaging and markings. Unless otherwise specified (see 6.2), six gas generators packaged and marked in accordance with Drawing 3205AS199 should be packaged into a fiberboard box conforming to ASTM D5118/D5118M, type CF or SF, class weather-resistant and WWVR, Style RSC. Gas generators for shipment to the Navy should be packaged and marked in accordance with Drawing 3205AS199. Fiberboard containers should have no single dimension, e.g., length, width, height, exceeding 45 inches. A tight pack should be mandatory. (Note: the hazard classification and National Stock Number (NSN) for the gas generator, has been

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established based on performed testing in container per Drawing 3205AS199. Any changes in approved container may require hazard classification testing and assignment of a new NSN.)

6.5.1 Normal marking. Unless otherwise specified (see 6.2), the marking on the outer fiberboard box and palletized unit loads should be as specified below. The specified markings should be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

Marking

a. NSN: 1377-01-448-9910

b. DoDIC: SS49

c. Generator, Gas, Dry Bay Fire Extinguisher GGU-15A/A

d. Drawing 838AS243

e. Quantity

f. Gross weight and cube

g. Lot number in accordance with MIL-STD-1168

h. Contract or purchase order number

6.5.2 Barcoding. Unless otherwise specified (see 6.2), the outer box and palletized load should be barcoded in accordance with MIL-STD-129.

6.6 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28 should be used to clarify terms.

6.7 Subject term (key word) listing.

Fire suppression system
Initiator
Propellant

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2024-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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